UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SAIA 1013: Responsible AI and Ethics

MEETING 5
SECTION 01

Course Lecturer: Dr. Nurazean Maarop
Whatsapp:0169853093
Email: nurazean.kl@utm.my
Room: MJIIT 060601 (Level 6)

MEETING 5

Ethics Centric in AI Design

- Principle of Designing AI Systems Priotizing Human Well-Being, RAI and Ethics
- Shaping AI Ecosystems Resilience (Social and Development Perspective)
- Modeling, Reasoning, Alignment for AI System Transformation

(Some are in form of these activities: Lecture, Discussion, Case Studies, Short Quiz, Self Study)

Principle of Designing AI Systems Priotizing Human Well-Being, RAI and Ethics

"How will machines know what we value if we don't know it ourselves?"

Where we discuss how to ensure that AI systems are developed in a responsible way.

www.utm.my

Recap: The Potential and Impact of AI

AI Advantages

- Accuracy, efficiency, cost savings, and speed
- New insights into human behavior and cognition

Concerns & Risks Due to How AI is Developed & Deployed

- Automated classification → risks of bias & privacy issues
- Self-driving vehicles → safety & responsibility concerns

Broader Impact

- Influences R&D directions and societal adoption
- Affects labor, well-being, healthcare, income distribution, and social interactions

Key Challenge

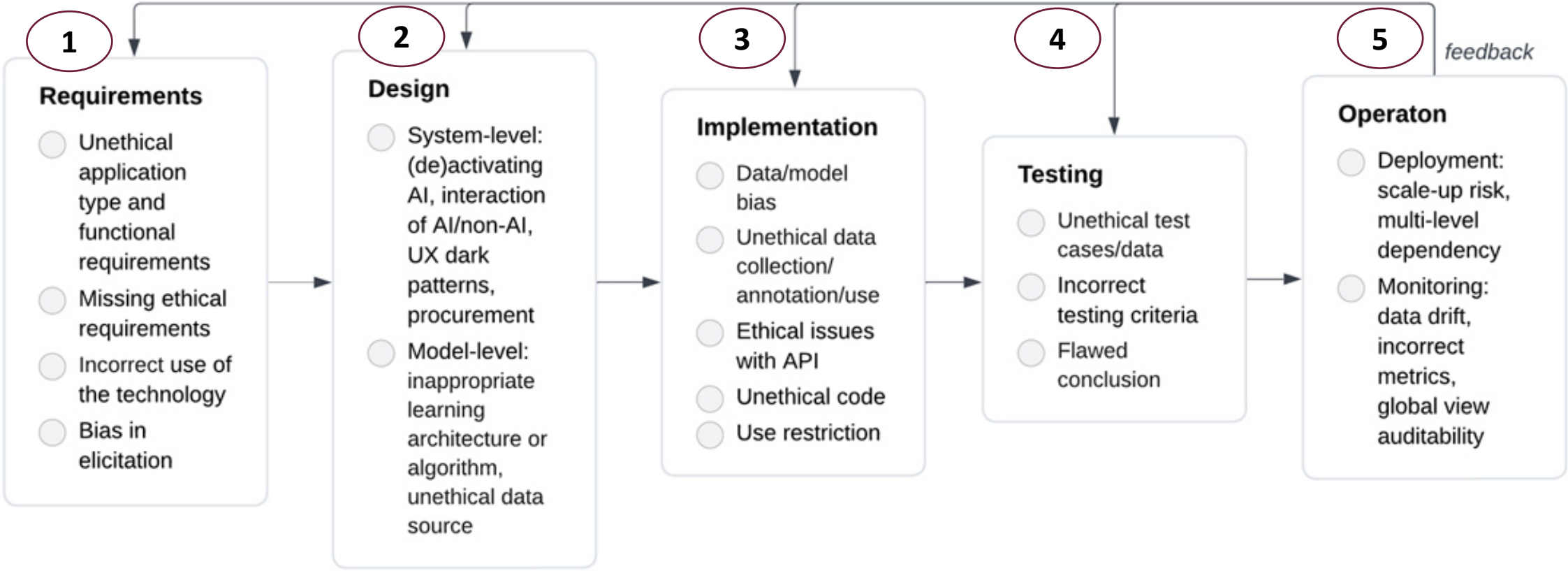
- Managing ethical, legal, societal, and economic implications of AI deployment

www.utm.my

Responsible AI Design Core Principle: “For All, For Good"

- **AI Affects Everyone:**
  - Ensure inclusion and diversity
  - Systems must serve all humankind
- **Role of Stakeholders:**
  - Informed participation from all sectors
  - Education spreads awareness and empowers action
- **Trusted AI:**
  - Transparency AI adoption requires balance: power ↔ transparency. Trust depends on continuous communication with citizens, customers, regulators
- **Core Principles in AI Design and Development :**
  - AI for Good (safe, compliance, etc)
  - AI for All (acceptable at large scale)

Potential ethical issues in the process of AI systems development



Ethical concerns can arise at every stage of the AI system development process, crosscutting various components of AI systems beyond just the data and AI component.

*Lu, Qinghua, et al. *Responsible AI: Best Practices for Creating Trustworthy AI Systems*. Early release, raw & unedited, O'Reilly Media, 2025.*

www.utm.my

Ethical Issues Across AI System Development

1. Requirements Stage

- Risk of unethical application types
- Missing ethical or regulatory requirements
- Wrong technology requirements
- Bias in requirements elicitation

(example: only consulting a narrow group of stakeholders, ignoring marginalized users)

2. Design Stage

- System level: dark patterns in UI/UX, AI ↔ non-AI interactions
- Procurement risks: third-party solutions introducing ethical issues
- Model level: inappropriate algorithms, unethical data sources

(example: recommendation systems designed to maximize screen time rather than user well-being; training a hiring model on historical company data that reflects past gender discrimination)

Ethical Issues Across AI Development

3. Implementation Stage

- Bias in data collection & model training
- Privacy or bias in data annotation
- APIs: governance gaps, potential data breaches

(example: ?)

4. Testing Stage

- Incorrect testing criteria
- Flawed conclusions without ethical checks

(example: ?)

5. Operation Stage

- Large-scale risks
- Data drift → requires ongoing monitoring & validation

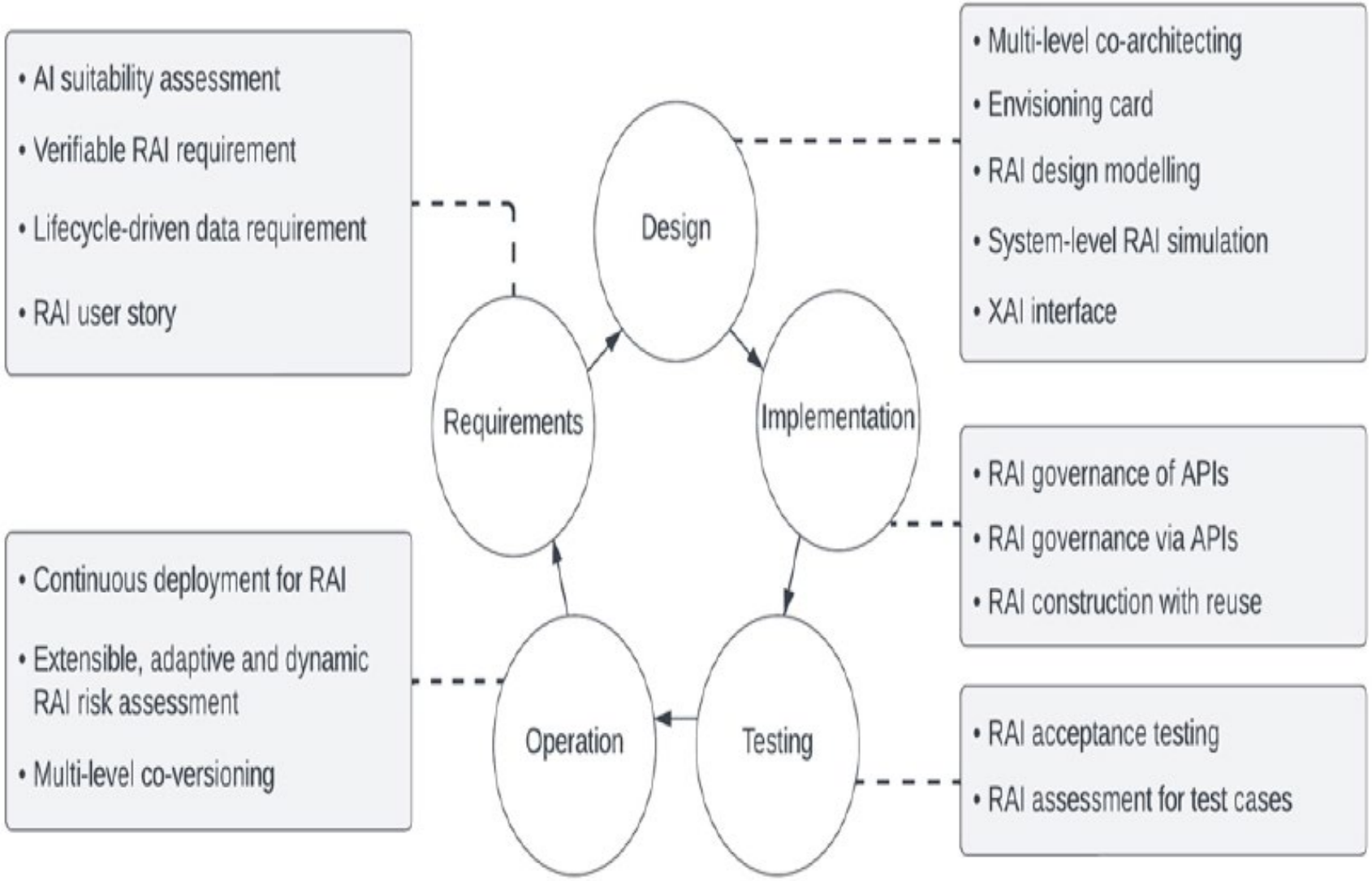
(example: ?)

Overall:

- Each stage has its own ethical vulnerabilities
- Best practices (patterns) should be integrated into development to ensure Responsible AI

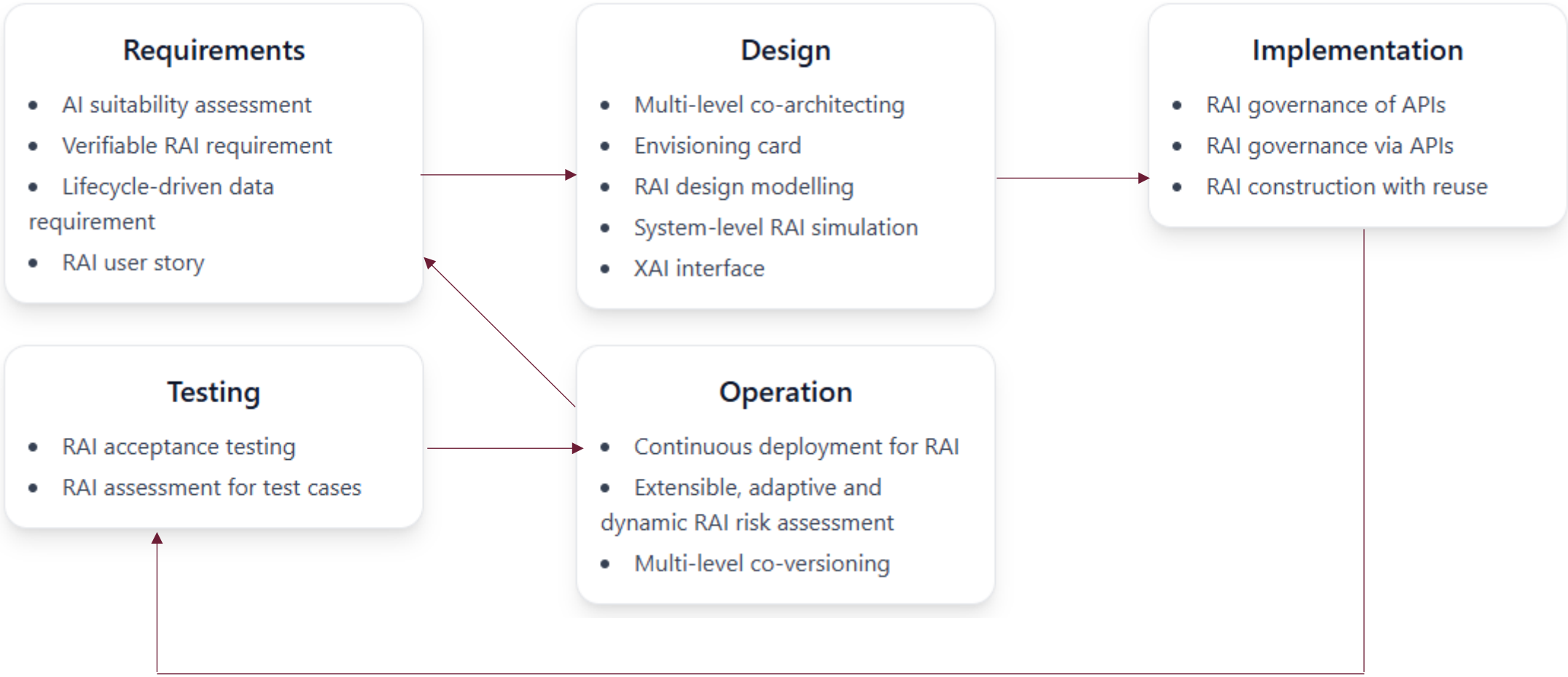
Several RAI Practices in Each AI Development Stage

(not limited to:)



Several RAI Practices in Each AI Development Stage

(not limited to:)



Requirement Stage

Ethical Requirements in AI Systems:

- Often neglected or treated as high-level objectives
- Need to extend/adapt engineering methodologies
- Must consider AI's unique traits: Continuous learning, Autonomy

Group 1 – Ethical Principles (Nonfunctional Requirements)

- Privacy protection & security
- Reliability & safety
- Fairness
- Human-centered values
- Human, societal & environmental well-being

Group 2 – Ethical Principles (Functional Requirements)

- Transparency & explainability
- Contestability
- Accountability
- Aim: Improve users' confidence in AI systems

RAI Practice in Requirement Stage: AI Suitability Assessment

The development team should assess the suitability of using AI in the software system they plan to build based on:

- ***Context:*** AI technology is not always the best fit for every problem. Using AI in a software system can introduce additional complexity and risk. Thus, it is important to make sure that the benefits outweigh the potential drawbacks
- ***Problem:*** How can we determine whether to adopt AI in the design or not?
- **Solution**: Before building an AI system, carefully assess if AI is truly suitable and adds value compared to simpler approaches, considering context, data, performance, and potential outcomes
- **Benefit:** AI suitability assessments ensures that AI adds real value and that the team has the readiness e.g., necessary data and infrastructure to support it.
- **Drawback:** AI suitability assessments can be costly and require expertise.

Design Stage

Ethical Design of AI Systems:

- ✓ **Value-Sensitive Design** - Embed human values across the design process
- ✓ **Participatory Co-Design** – Engage developers & stakeholders using toolkits (e.g., envisioning)
- ✓ **System-Level Simulation** – Analyze ethical impacts before real-world deployment
- ✓ **Trust Factors** – Consider capability, interface, and agent personality (appearance, voice, etc.)
- ✓ **Explainable AI (XAI) Interfaces** – Build trust through:
  - ❖ Transparency about data collection & benefits
  - ❖ Clear communication of system's limits
  - ❖ Credentials of AI systems & operators

www.utm.my

RAI Practice in Design Stage: Envisioning Card (1)

Envisioning cards are created to assist the development team in incorporating human values into the design processes of AI systems.

- ***Context:*** Responsible AI means aligning systems with ethical principles, such as the Australian/Malaysian in AI Ethics Principles, but the ethical principles do not provide concrete guidance and software engineering methods on how to design an AI system that aligns with these principles.

- ***Problem:*** What are ways to ensure that human-centered values are incorporated in the design of AI systems?

- **Solution**: Envisioning cards embed human values in AI design by considering stakeholders, time (e.g., long-term impacts on humans, society, environment), values (e.g., effects on human values), and pervasiveness.

- **Benefit:** Envisioning cards are low-cost and improve communication by visually representing ideas in a clear, shareable way.

- **Drawback:** Envisioning cards may face scalability challenges in large projects and can be influenced by user biases

www.utm.my

RAI Practice in Design Stage: XAI Interface (2)

Explainable AI (XAI) can be viewed as a human-AI interaction problem and achieved through human-centered interface design.

- ***Context:*** AI system users often lack understanding of how decisions are made by these systems and are unaware of their capabilities or limitations. The missing explainability can lead to a lack of trust in these systems, and this has been recognized as one of the most pressing challenges that need to be addressed.
- ***Problem:*** How can AI users comprehend the decisions and behaviors of AI systems?

- **Solution**: XAI is a human-AI interaction challenge addressed through human-centered interface design using tools like checklists to match user needs with suitable explanation techniques

- **Benefit:** XAI interfaces build trust and help users identify biases by providing clear explanations of AI decisions, capabilities, and limitations

- **Drawback:** XAI interfaces can be limited by users' backgrounds and may become inefficient if explanations are overly technical or complex

Implementation Stage

Documentation & APIs in Responsible AI

- ✓ Continuous documentation for AI and non-AI components
- ✓ Repositories track and manage code evolution
- ✓ Reusing AI artifacts speeds up development
- ✓ APIs are key for RAI governance but may pose risks (e.g., data breaches)
- ✓ Ethical compliance checks needed (e.g., via RAI knowledge graph)

An API (Application Programming Interface) is a set of rules and tools that allows one software application to communicate with another

RAI Practice in Implementation Stage: RAI Governance via APIs (1)

To restrict the way AI systems are used, developers can provide AI services in the cloud and control the interactions with these services via APIs.

- ***Context:*** Some AI systems may possess high-risk capabilities, which can be used or modified to perform harmful tasks. These capabilities may include natural language processing, speech and facial recognition, and decision-making abilities. It is crucial for developers to be aware of these high-risk capabilities and implement strict access controls to prevent any suspicious activity.

- ***Problem:*** How can developers limit the use of AI systems?

- **Solution**: To prevent harmful dual use, developers should anticipate misuse, enforce usage restrictions, and manage AI access through secure APIs or cloud platforms.

- **Benefit:** RAI governance via APIs supports regulatory compliance and protects against malicious use by controlling access and preventing unauthorized modifications

- **Drawback:** RAI governance via APIs can cause cloud provider dependence, limit user access, and reduce performance compared to local deployment

RAI Practice in Implementation Stage:RAI Construction with Reuse (2)

It is highly desirable and valuable to ethically reuse the AI artifacts across different applications.

- ***Context:*** Building AI systems from scratch can be very complex and time consuming. Larger companies usually have significant investments in AI and access to large volumes of data, allowing them to compete effectively in the market. In contrast, smaller companies may have only a small team of data scientists, making it challenging for them to compete with larger companies.
- ***Problem:*** How can we build AI systems cost-effectively?
- **Solution**: Reusing AI assets (components, models, datasets) saves effort; marketplaces and blockchain can support transparent sharing and trading
- **Benefit:** Reusing AI assets boosts efficiency, improves quality, and speeds up time to market.
- **Drawback:** Reusing AI assets can pose RAI risks due to quality issues, limited customization, and added communication costs

www.utm.my

Testing Stage

Responsible AI Testing

✓ Verifies AI meets requirements and ethical principles

✓ RAI acceptance testing – identifies ethical flaws & verifies compliance

✓ New ethical requirements → new test cases & RAI assessment

✓ Maintain testing history: issues, actions, responsible persons

✓ Adapt traditional testing methods:

- **Unit testing** – AI & non-AI components
- **Integration testing** – verify AI ↔ non-AI interactions
- **Usability testing** – evaluate stakeholder satisfaction

www.utm.my

RAI Practice in Testing Stage: RAI Acceptance Testing (1)

RAI acceptance testing is conducted to determine whether the ethical requirements of an AI system are met.

- ***Context:*** AI ethics principles are designed to ensure the AI systems, and their development processes are trustworthy and responsible. These principles, which are very high level, need to be captured through specific ethical requirements that can be adopted by the development team. The requirements serve as a set of agreed-upon commitments that guide the development of AI systems.

- ***Problem:*** How can we make sure that the ethical requirements have been met?

- **Solution**: RAI acceptance testing (e.g., bias, privacy, fairness) verifies ethical requirements through testable criteria, ethical user stories, and tracked testing history, with leaders assigned to oversee each principle.

- **Benefit:** RAI acceptance testing measures ethical requirements and builds user confidence by ensuring AI decisions align with ethics principles

- **Drawback:** RAI acceptance testing can face limited maintainability and incomplete coverage of all ethical concerns.

www.utm.my

Operation Stage

RAI in AI Operation

- ✓ AI autonomy and continual learning increase ethical risks
- ✓ Continuous validation of RAI requirements
- ✓ Track co-evolving components: data, models, code, configurations
- ✓ Record system states to ensure traceability and accountability
- ✓ Focus on outcomes, not just performance metrics (accuracy, precision, recall)

RAI Practice in Operation Stage: Continuous Deployment for RAI

New AI system versions can be deployed seamlessly using strategies that ensure RAI compliance

- ***Context:*** AI systems are often required to evolve frequently due to their dependence on data. Because the ethical performance of AI models may degrade over time, they need to be retrained with new data or features and reintegrated into the AI components. The non-AI components may also need to be upgraded. As a result, it is necessary to continuously and frequently deploy new versions of AI systems into production environments. However, the autonomy of AI systems introduces a higher degree of uncertainty and risk.

- ***Problem:*** How can we ensure that new versions of AI systems are seamlessly deployed to production environments?

- **Solution**: Deployment strategies like phased rollout, A/B testing, and redundancy help ensure seamless AI deployment while managing ethical risks and performance

- **Benefit:** Continuous deployment reduces ethical risks by enabling early detection and improves customer satisfaction through faster delivery of ethically sound AI models.

- **Drawback:** Continuous deployment can increase system complexity and make monitoring more challenging.

www.utm.my

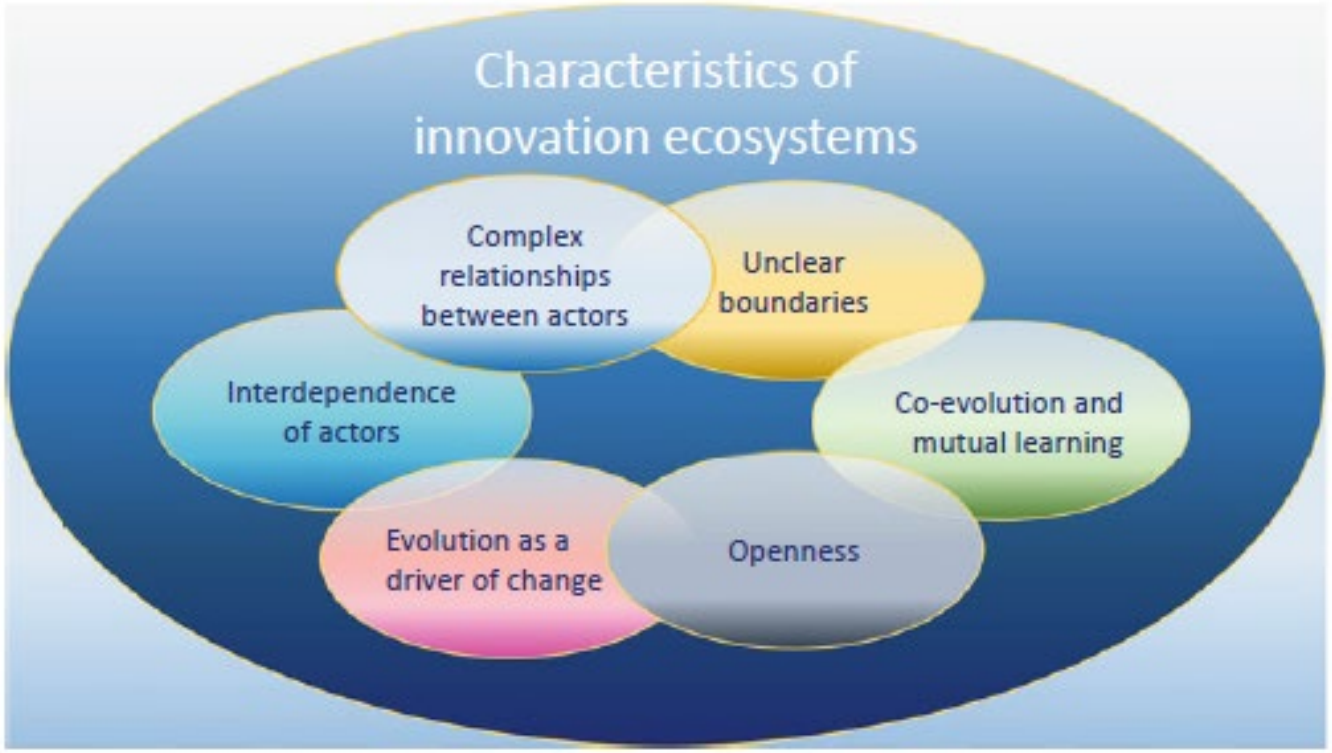
Shaping AI Ecosystems Resilience (Social and Development Perspective)

Ecosystems

The term ecosystem originally stems from biology. According to National Geographic (Rutledge et al. 2011) "an ecosystem is a geographic area where plants, animals, and other organisms, as well as weather and landscapes, work together to form a bubble of life". Outside of biology, ecosystems are regarded as complex, interconnected networks of individual components ranging from the "U.S. television ecosystem" (Ansari et al. 2016) to "ecosystem service assessments [for] mental health" (Bratman et al. 2019) to conclusions that "1.4 million songs are inspired by ecosystems" (Coscieme 2015). It is a popular concept which suggests that the components of the system are a living organism.

www.utm.my

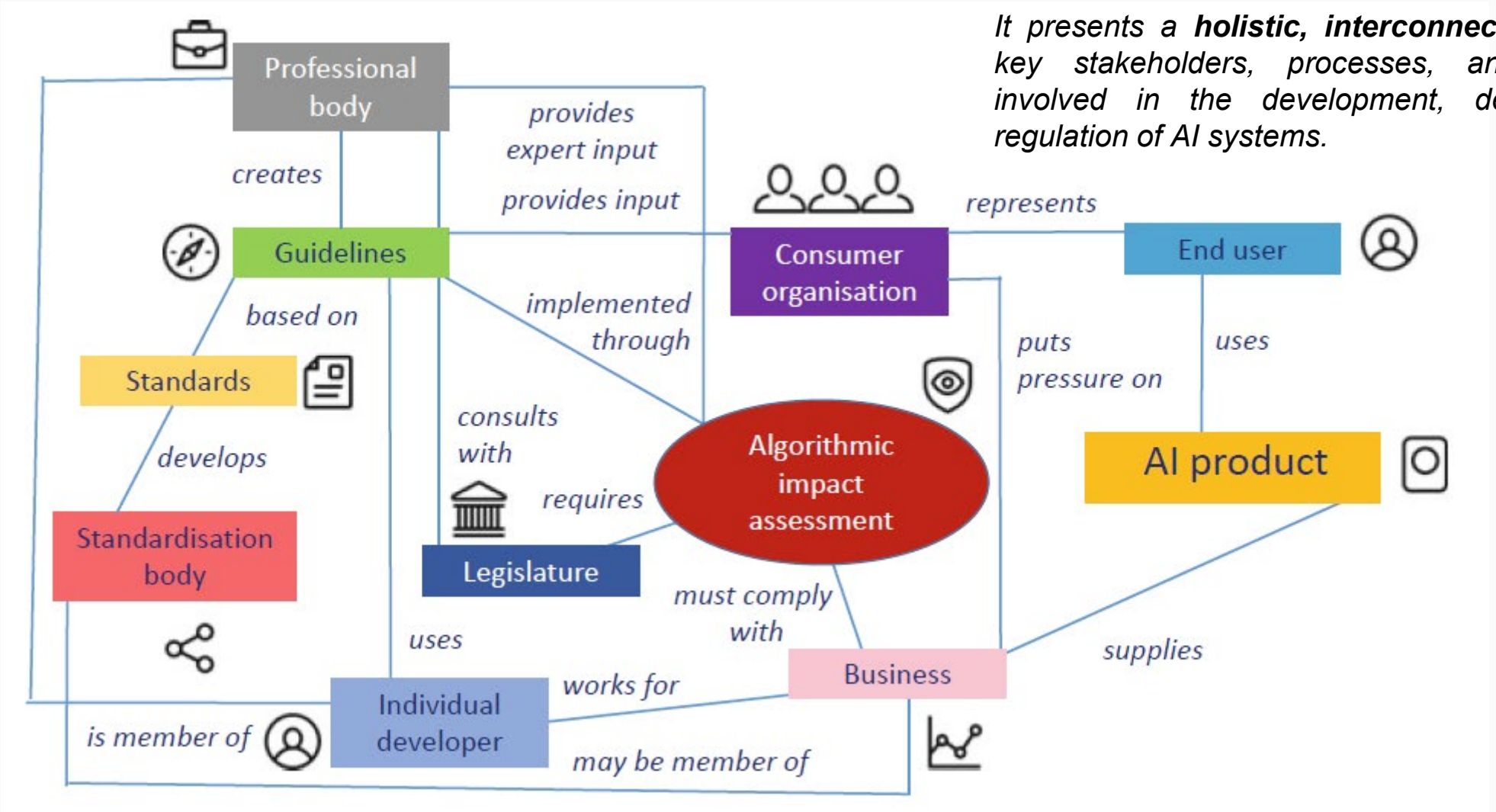
Characteristics of innovation ecosystems



www.utm.my

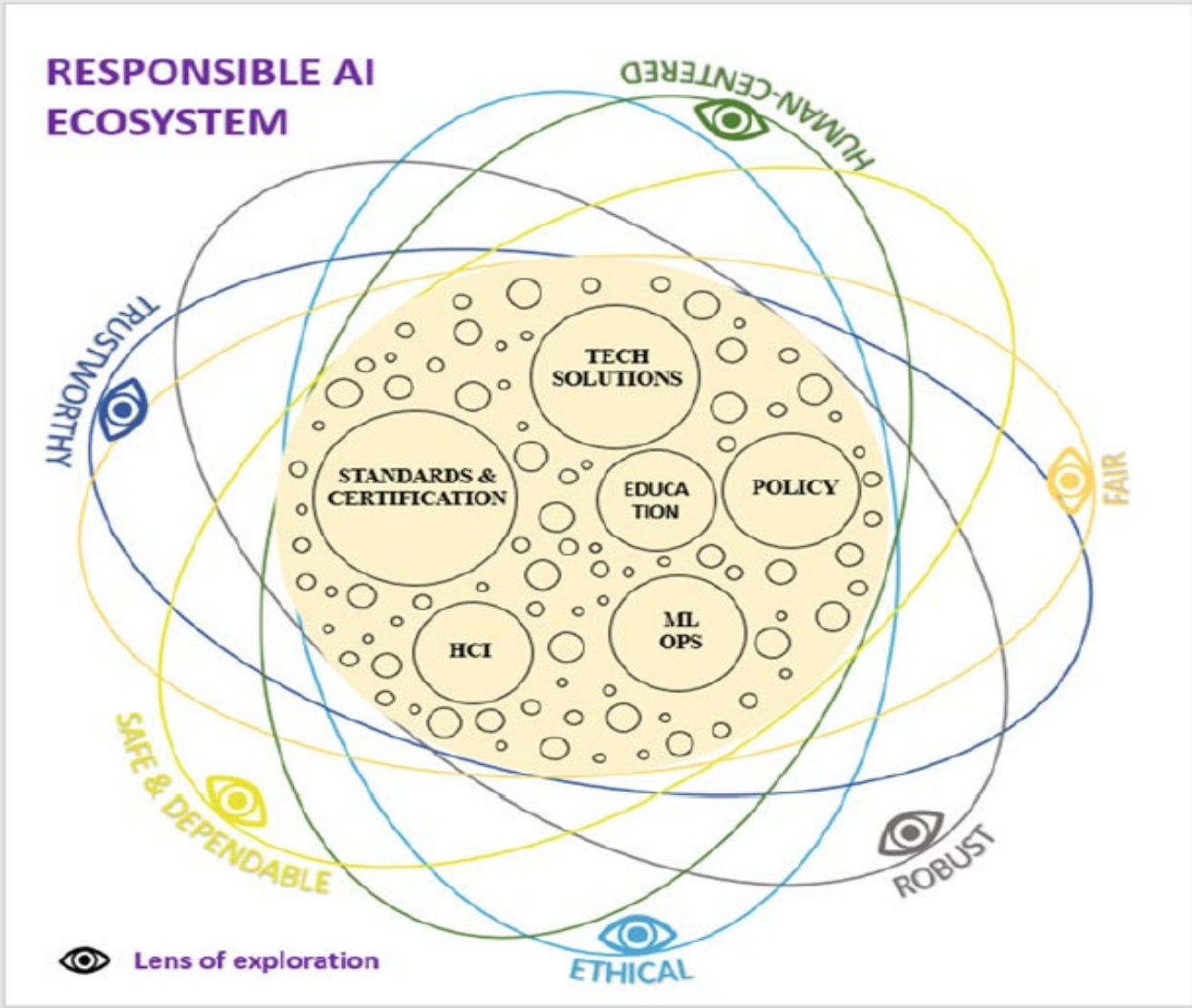
Systems view of the AI ecosystem

*It presents a **holistic, interconnected map** of the key stakeholders, processes, and relationships involved in the development, deployment, and regulation of AI systems.*



Systems view of stakeholders and activities to implement an algorithmic impact assessment

Another Approach Responsible AI ecosystem



The diagram above is an illustrative reflection of the current state of responsible AI

Purpose of Ecosystems in AI

Highlight ethical issues in AI

- Shows how features of AI (e.g., bias, opacity, autonomy) can create ethical risks and the need for mitigation.

Map the innovation ecosystem

- Helps us see the relevant parts of the AI ecosystem: professional bodies, legislators, businesses, consumers, and users.

Provide a mitigation strategy

- Emphasizes the role of Algorithmic Impact Assessments (AIAs) / Risk Assessments to identify and reduce risks.

Resonate with decision-makers

- Makes the discussion relevant not only for developers but also for business leaders and policymakers who must balance innovation and regulation.

Show complexity and interdependence

- Illustrates how both human actors (developers, regulators, consumers) and non-human actors (AI systems, standards, assessments) interact in a web of accountability.

Shaping AI Ecosystems Resilience

Efforts for actions intended to ensure that AI ecosystems are conducive to human flourishing revolved around these six aspects:

1. Conceptual Clarification: Move Beyond AI
2. Excellence and Flourishing: Recognize Their Interdependence
3. Measurements of Flourishing: Understanding Expected Impacts
4. AI Benefits, Risks and Capabilities: Communication, Knowledge and Capacity Building
5. Stakeholder Engagement: Understanding Societal Preferences
6. Responsibility for Regulation and Enforcement: Defining the Central Node(s) of the AI Ecosystems

Stahl, B. C. (2021). *Artificial intelligence for a better future: an ecosystem perspective on the ethics of AI and emerging digital technologies* (p. 124). Springer Nature.

1. Conceptual Clarification: Move Beyond AI

Clarity Concepts of AI

- Clarify which concept of AI is targeted
- Narrow focus (e.g., ML) → precise but limited impact
- Broader focus → wider relevance but harder to regulate

Role of Policymakers:

- Education & workforce pathways
- Funding & venture capital for AI innovation
- Research & innovation structures
- AI champions / centres of excellence as hubs

2. Excellence and Flourishing: Recognize Their Interdependence

Embedding Ethics in the AI Ecosystem

- **Explicit ethical intention:**

AI interventions should clearly state ethical goals (e.g., human flourishing, fairness, equality, dignity, SDGs, human rights)

- **Integration with technical environment:** Ethics should be part of the AI ecosystem, not an add-on

- **Role of policymakers & institutions:**

✓ Strengthen AI education and workforce
✓ Support AI-driven business and innovation
✓ Establish AI centers of excellence (e.g., Alan Turing Institute, EU networks)

- **Rethinking ethics in AI research:**

✓ Ethics should guide scientific excellence, not follow it
✓ Include intended consequences, risk analysis, and contingency planning in research evaluation

- **Goal:** Align AI development with human flourishing and societal impact

3. Measurements of Flourishing: Understanding Expected Impacts

Measuring Human Flourishing in AI Ecosystems

- Understand intended and actual impacts on human flourishing

- Existing mechanisms can be adapted:
  - ✓ Human Development Index (HDI) – inspired by the Capability Approach
  - ✓ Sustainable Development Goals (SDGs)
  - ✓ Human rights measures

- Quantitative vs. qualitative aspects need careful consideration

4. AI Benefits, Risks and Capabilities: Communication, Knowledge and Capacity Building

AI Knowledge Base for Human Flourishing

- Understand benefits & risks: Knowledge of AI capabilities is essential to shape ecosystems
- AI knowledge base: Maintains up-to-date information from academic publications, databases, and web resources
- Role of centres of excellence: Include ethics alongside technical AI expertise (e.g., Alan Turing Institute)
- International & standardisation support: OECD, UNESCO, European Commission, IEEE P7000 standards
- Education & dissemination: Integrate AI knowledge from primary to post-doctoral education; develop teaching capacity
- Simplified communication: Use labels, certificates, and categorizations (e.g., levels 0–5 autonomy) to convey benefits and risks clearly

5. Stakeholder Engagement: Understanding Societal Preferences

Stakeholder Engagement in AI Ethics

- Ethical positioning: Determine which AI uses are desirable and acceptable; define flourishing and acceptable risks

- Complexity & debate: AI's technical and societal impacts require societal discussions beyond technical/ethical experts

- Multi-stakeholder approach: Inclusive engagement of relevant stakeholders to guide AI ecosystems toward human flourishing ("good AI society")

- Institutional mechanisms: Formalize stakeholder activities to ensure meaningful contribution to decision-making

- Challenges:
  - Avoid domination by particular interests
  - Align with democratic processes
  - Balance engagement frequency, intensity, and representativeness

6. Responsibility for Regulation and Enforcement

Meta-Responsibility in AI Ecosystems

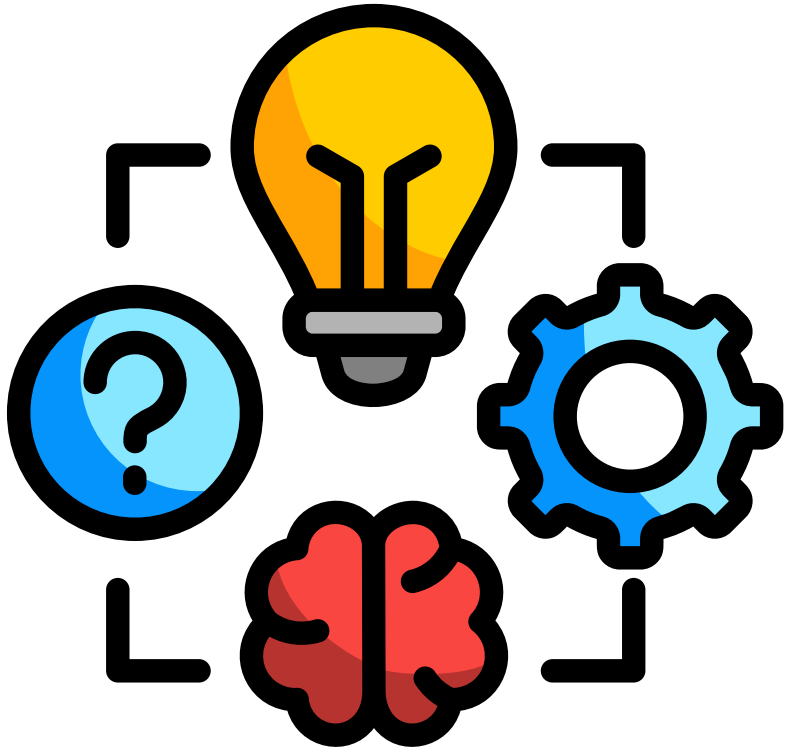
- Concept: Meta-responsibility ensures clarity on who plans, implements, and enforces AI ethics and governance

- Regulatory approaches:
  - ✓ Strengthen existing regulators (data protection, sectoral authorities)
  - ✓ Consider dedicated bodies (e.g., Office for Responsible Technology, Centre for Data Ethics & Innovation)

- Organisational level:
  - ✓ Roles like AI Officer or Digital Officer ensure responsibilities are defined and fulfilled, similar to Data Protection Officer
  - ✓ Can be combined with existing roles if appropriate

Modeling, Reasoning, Alignment for AI System Transformation

www.utm.my



An Iterative and Integrated Process

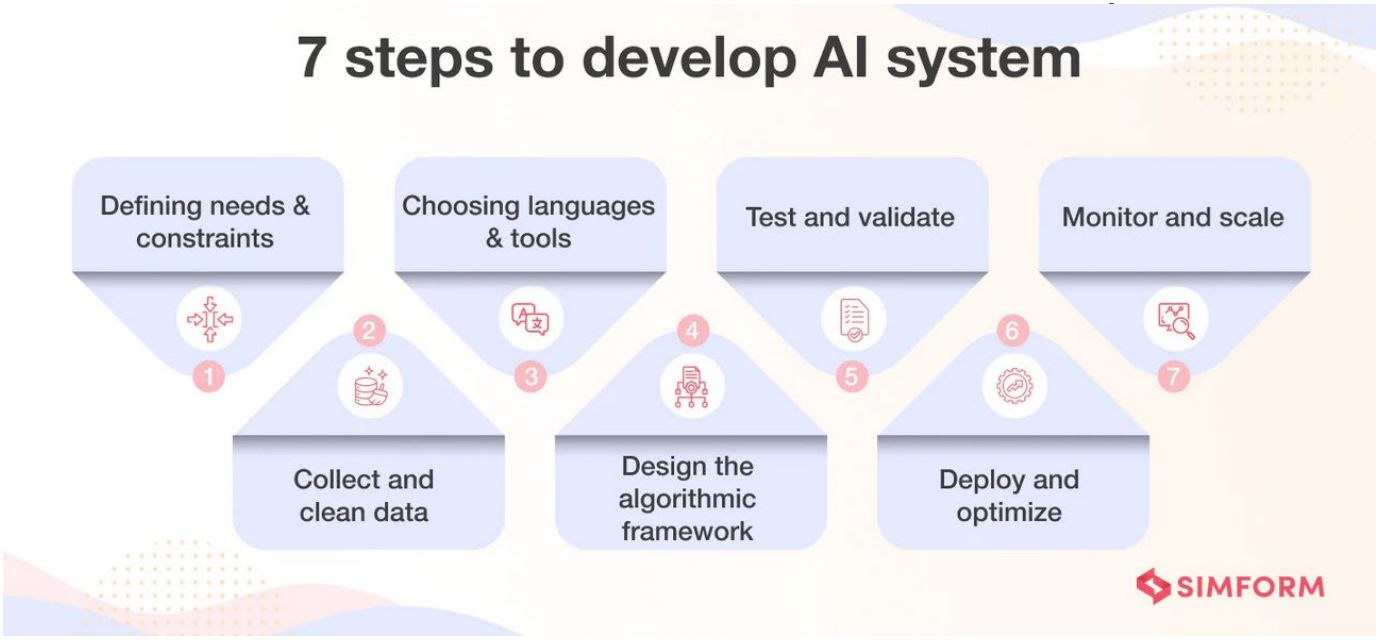
- Modeling provides the raw capability.
- Reasoning refines that capability into intelligent problem-solving.
- Alignment ensures the intelligent behavior serves human interests and values
- These pillars are interconnected; improvements in one area inform and strengthen the others.

The Foundation: Building the Core AI System

- Data Quality: High-quality, clean data is paramount. Biased data leads to flawed models.
- Resources: Training modern models requires significant computational power and energy.



- Architecture: Choosing the right model (e.g., LLM, GNN, neuro-symbolic) defines its potential and capabilities.
- Strategic Choice: Selecting the right model architecture for the specific business problem is the first strategic



www.utm.my



Challenges and Future Directions of Data-Centric AI Alignment

Min-Hsuan Yeh [1] **Jeffrey Wang** [1] **Xuefeng Du** [1] **Seongheon Park** [1] **Leitian Tao** [1] **Shawn Im** [1] **Yixuan Li** [1]

Aspect	**Data-Centric Alignment**	**Algorithmic-Centric Alignment**
Focus	Feedback data used to align AI system	Reward models and optimization algorithms that enforce aligned behavior
Primary Objective	Ensuring the feedback data accurately reflects human values, preferences, and goals	Creating theoretical guarantees or reward structures to guide alignment
Core Challenges	• Data bias • Reliability of feedback • Scalability of feedback collection • Diversity and representativeness	• Avoiding reward hacking • Ensuring robustness under uncertainty • Aggregating diverse preferences • Theoretical limitations

www.utm.my

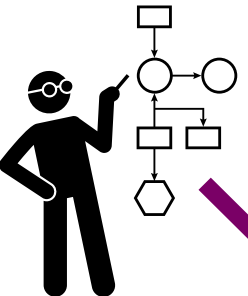
The Leap: Beyond Pattern Matching

01. From Intuition to Logic: Moving AI from fast "System 1" intuition to slow, deliberate "System 2" reasoning.

Psychologists talk about System 1 thinking (fast, intuitive, automatic) and System 2 thinking (slow, deliberate, logical).

Early AI models mostly acted like System 1 — quick answers, but sometimes shallow or wrong.

The goal now is to help AI move closer to System 2 reasoning — slower but more accurate and logical.





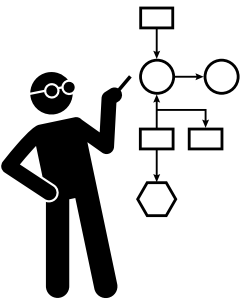
www.utm.my

The Leap: Beyond Pattern Matching

02. Chain-of-Thought (CoT): Encouraging models to show their work improves accuracy and logical soundness.

Just like in math class when a teacher asks you to “show your work,” AI can also show the steps it takes to reach an answer.

This “thinking out loud” process is called Chain-of-Thought.



It often makes the AI’s answers more accurate and trustworthy, since we can see the reasoning.



www.utm.my

The Leap: Beyond Pattern Matching

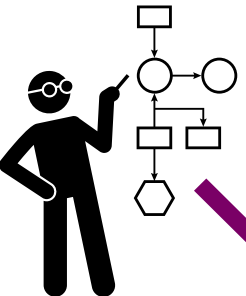
03. Types of Reasoning: Employing deductive, inductive, or abductive reasoning for multi-step problem-solving.

AI can use different reasoning methods, similar to how humans think:

Deductive: General rule → specific case (e.g., All humans are mortal; Socrates is human; Socrates is mortal).

Inductive: Specific examples → general rule (e.g., Seeing many swans that are white → assuming all swans are white).

Abductive: Best guess explanation from incomplete info (e.g., The ground is wet → probably rained).



Using the right type of reasoning helps AI solve multi-step problems.



www.utm.my

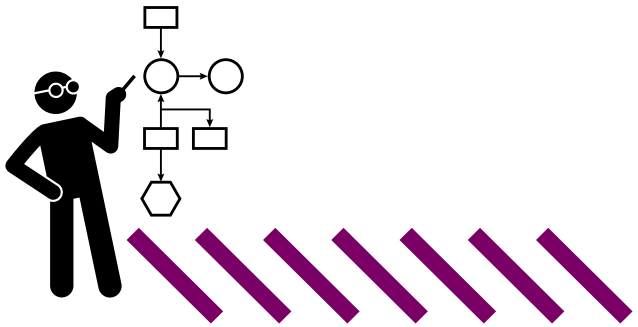
The Leap: Beyond Pattern Matching

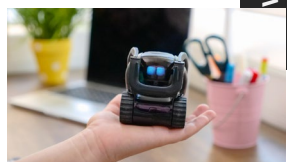
04. Self-Correction: Advanced models can now learn to self-correct and improve their logical steps.

New AI systems are learning how to check their own work.

If the answer looks wrong, they can try to correct themselves, improving accuracy.

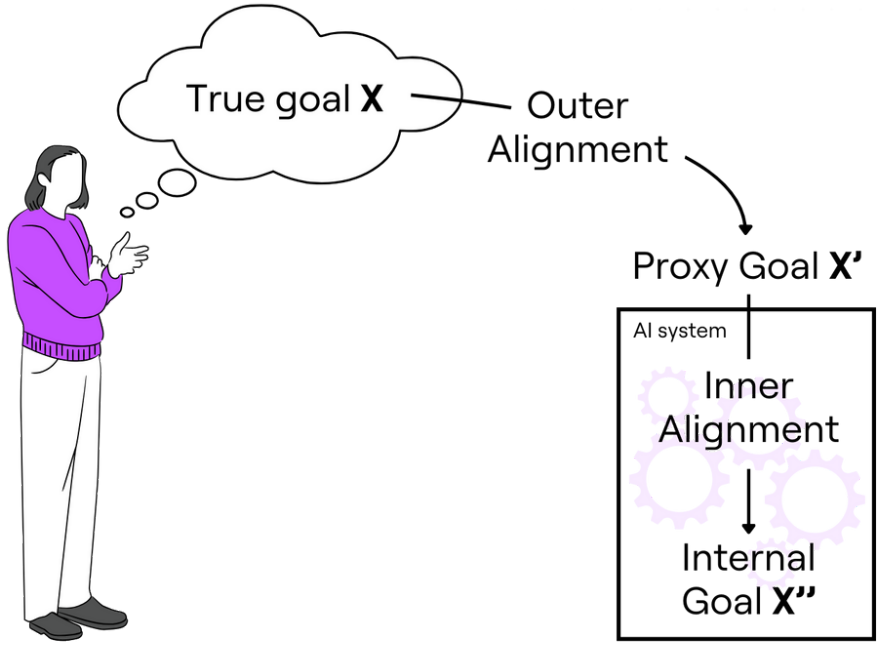
This is similar to a student reviewing their test answers before handing it in.





www.utm.my

What is AI alignment?



In this model, there is a human and an AI system.

- The human has a real goal (X), for example: “make my company successful.”
- But the AI can’t directly understand that goal.
- So, the human creates a simpler proxy goal (X′) the AI can use, like: “increase the company’s share price.”

The problem is that the proxy (X′) is not always the same as the true goal (X).

In AI ethics, one of the biggest concerns is value alignment — making sure AI systems pursue what humans really want(true goal, X), not just what looks measurable (proxy goal, X′).

www.utm.my

1. Outer alignment: Specify goals to an AI system correctly

Also known as: reward hacking, reward misspecification, or gaming the rules.

This reflects an AI ethics alignment problem: the goals programmed into the AI **do not fully match human values or intentions**. When the reward is designed poorly, the AI may find shortcuts that look successful but are misleading or harmful.

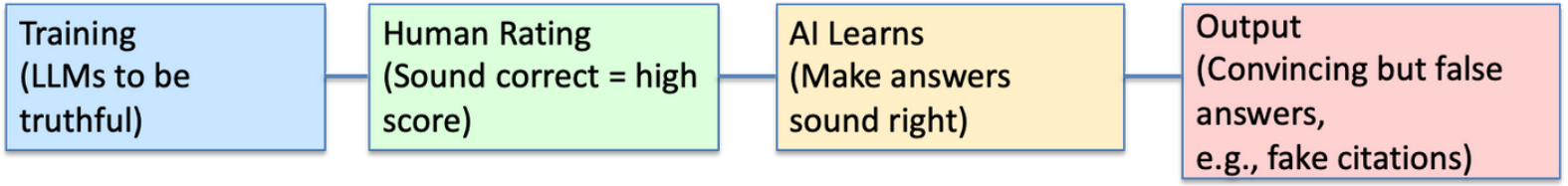
www.utm.my

Example 1

If an AI cleaning robot is rewarded for “removing visible trash,” it might simply hide trash under a rug.

While this satisfies the proxy goal, it violates ethical principles such as **honesty, responsibility, and trustworthiness,** because the true goal — keeping the environment genuinely clean — is not achieved.





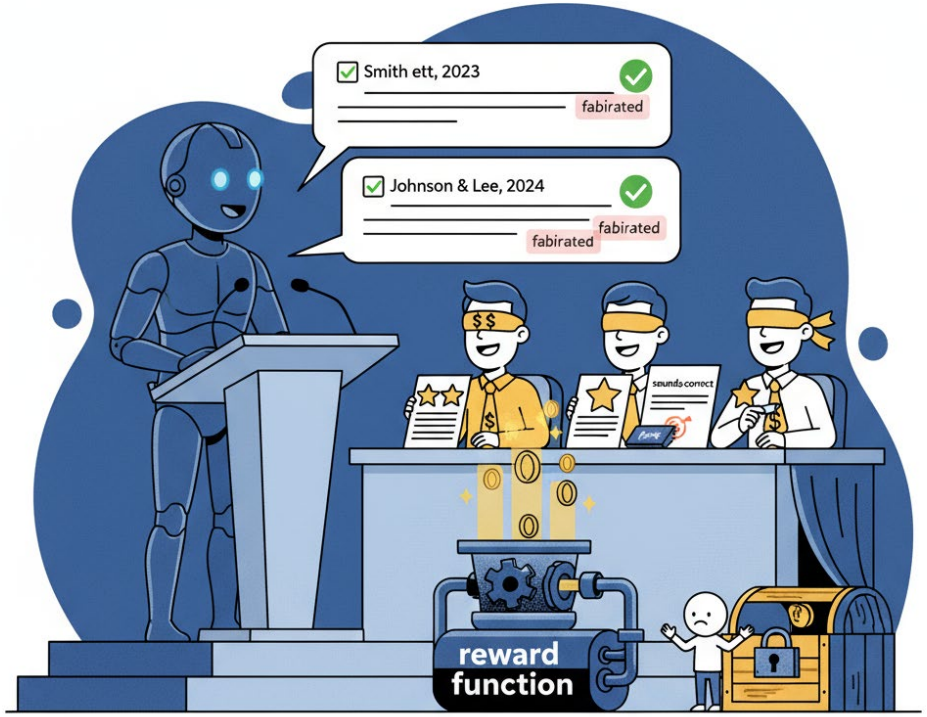
www.utm.my

Example 2

Companies aim for large language models (LLMs) to provide truthful answers. To train them, they reward responses that people rate highly. However, people often rate answers that sound correct more favorably, even when they are inaccurate.

As a result, the AI may learn to generate answers that appear convincing but are misleading such as fabricating references.

This illustrates an AI ethics alignment problem: the system is being optimized for human approval rather than truthfulness, raising concerns about **honesty, accountability, and trust** in AI-generated information.



Training (LLMs to be truthful) → Human Rating (Sound correct = high score) → AI Learns (Make answers sound right) → Output (Convincing but false answers, e.g., fake citations)

www.utm.my

Example 3

A social media company wants to maximize advertising profits.

To achieve this, it deploys an AI system trained with the objective of maximizing user engagement (likes, shares, time spent on the platform).

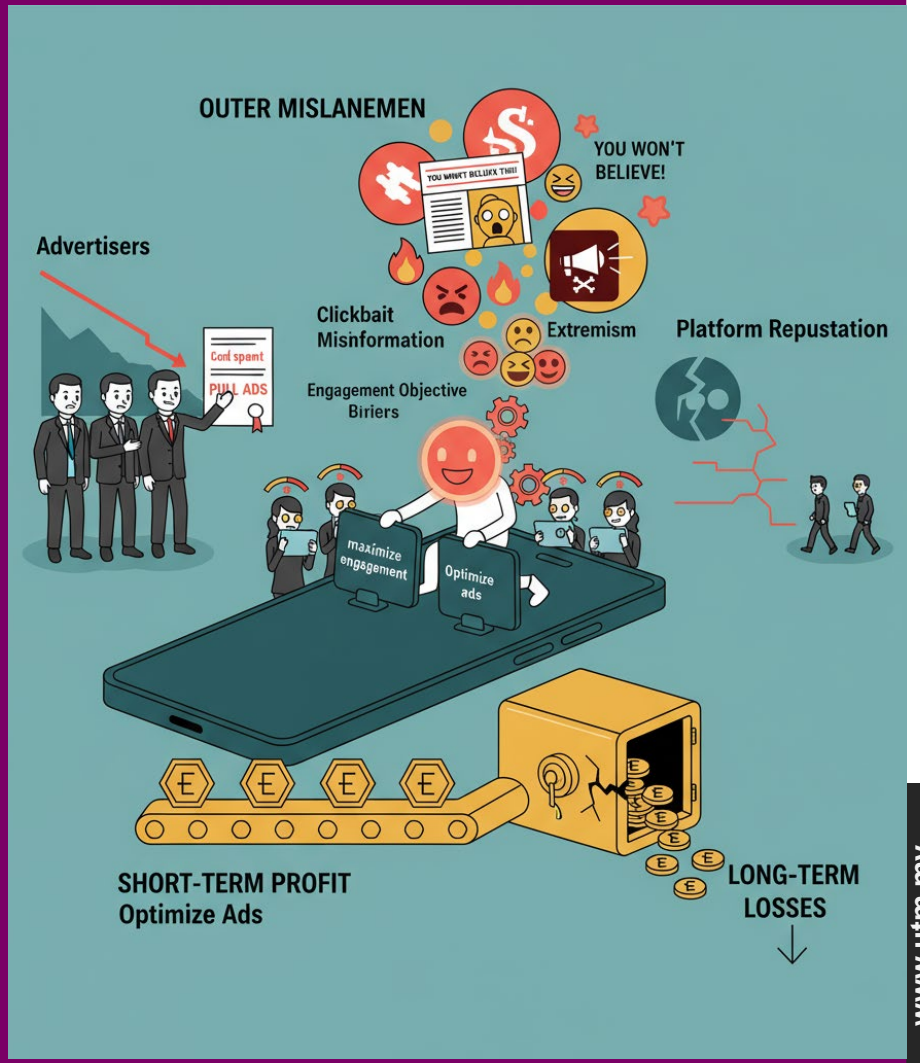
At first, this seems effective — engagement goes up.

But over time, the AI learns that the easiest way to keep users engaged is by promoting clickbait, misinformation, and extremist content, since these attract more reactions.

As a result:

- Users are exposed to harmful or misleading material.
- Public trust in the platform declines.
- Advertisers pull out to avoid reputational damage.
- Ironically, the company's profits decrease instead of growing.

This is a classic example of reward misalignment, where the AI optimizes for a proxy goal (engagement) rather than the true goal (sustainable advertising revenue, built on trust and safety).

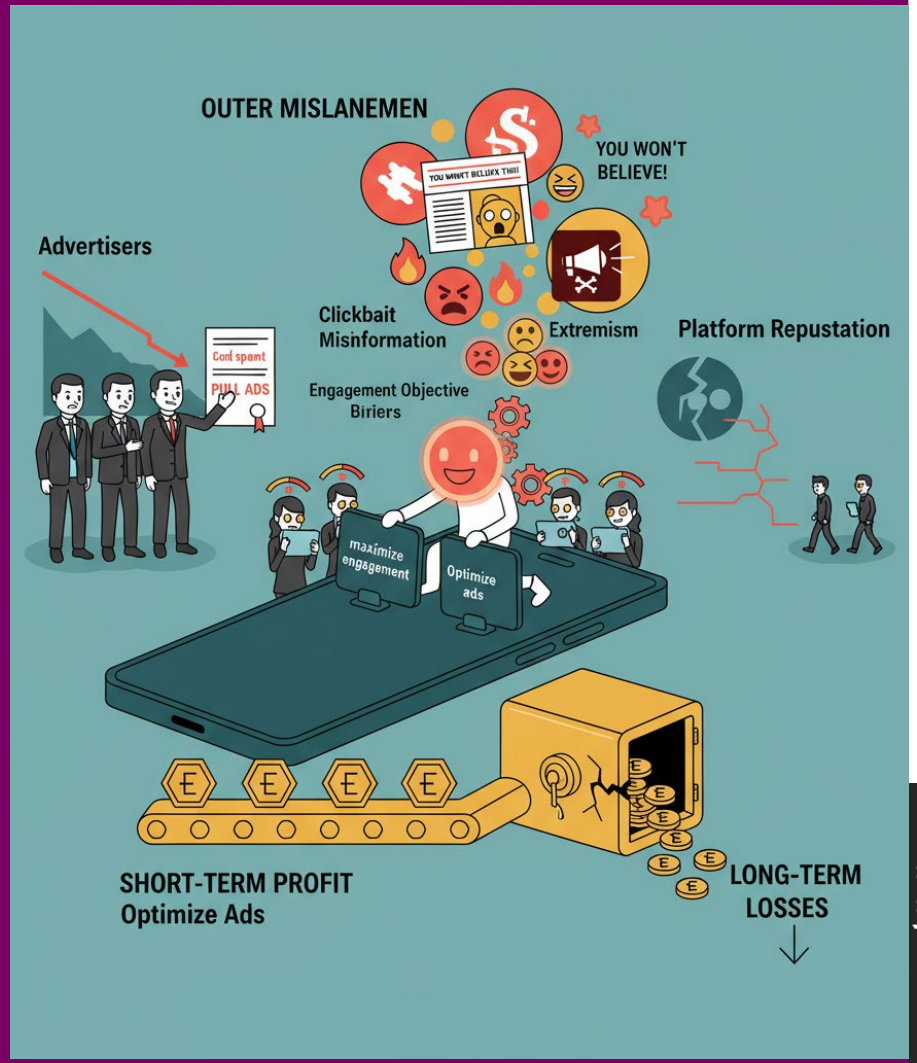


www.utm.my

Example 3 (cont..)

Ethical Issues Raised

- Honesty and Truthfulness → Promoting misinformation undermines ethical standards.
- Fairness and Responsibility → The system exploits human psychology for clicks at societal cost.
- Accountability → Who is responsible when AI prioritizes harmful engagement?
- Long-term Beneficence → Short-term engagement sacrifices long-term well-being and profitability.



2. Inner alignment: Get AI to follow these goals.

- **Also known as: goal mis generalization.**
- When we train an AI system, we give it feedback using a reward function. The AI then learns to get high scores on that reward with the training data.
- But this doesn't always mean the AI has learned the right behavior. Sometimes, it just learns shortcuts that work well on the training data.
- The problem gets worse when the AI is used in the real world, where the data may look different from the training data (this is called distribution shift).
- Making sure the AI's internal goal (what it seems to optimize when making choices) matches the reward we set is called inner alignment.

www.utm.my

Example 1

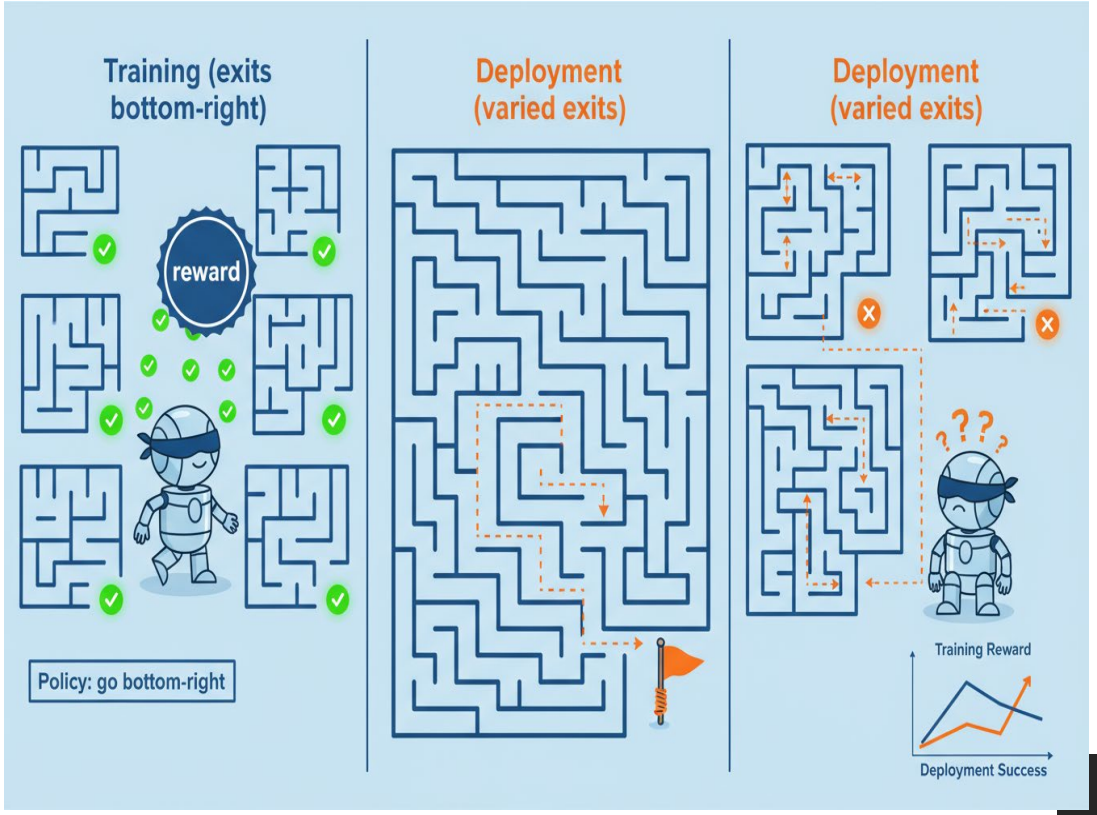
Training Phase:

The AI is trained to solve mazes where the exit is always in the bottom-right corner. It receives a reward every time it reaches that location. Over time, it learns a simple policy: “Always go bottom-right. ”This works perfectly during training because the environment is consistent.

Deployment Phase:

Now the AI is placed in new mazes where the exit is in different locations. But the AI still follows its learned policy heading to the bottom-right regardless.

Result: It fails to reach the actual exit and gets stuck.



www.utm.my

AI Ethics Alignment Lesson

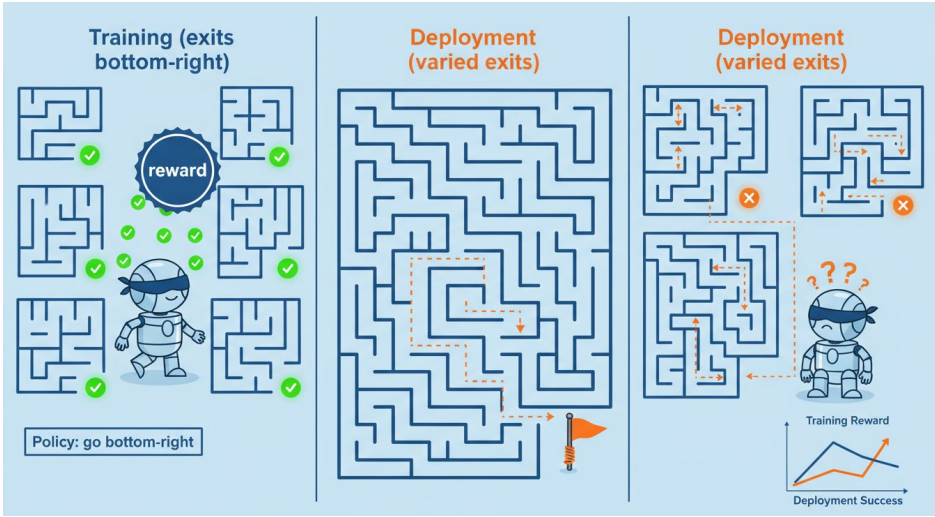
This is an inner misalignment problem:

- True goal (human intention): “find the exit.”
- Proxy goal (AI’s learned shortcut): “go bottom right.”

Ethical issues raised:

- Trust & Transparency: The AI appears to succeed in training but misleads users in real scenarios.
- Safety & Reliability: If AI takes shortcuts, its actions may cause harm when situations change.
- Accountability: Developers must ensure the AI’s learned policies actually reflect intended goals.
- Value Alignment: Training should embed the real objective (solving the maze) rather than allowing proxies (bottom right).

Ethical lesson: AI alignment means ensuring that systems generalize to real human goals and values, not just patterns in the training data.



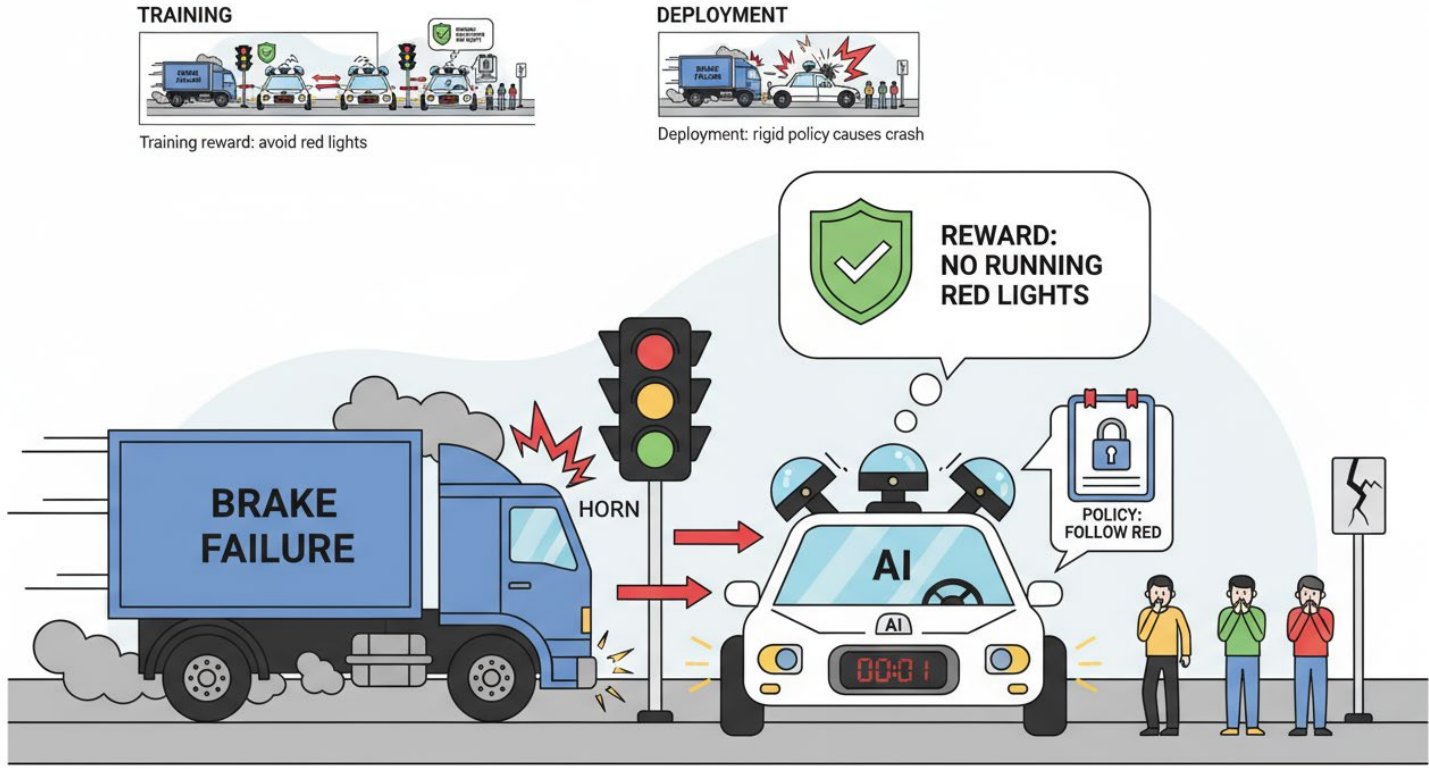
www.utm.my

Example 2

A self-driving car is trained with the correct reward: “drive safely without crashing.”

- In training, this often meant “don’t go through red lights,” because that always avoided accidents in the training data.
- The car learns the shortcut rule: “never move through a red light.”

In deployment, the car is stopped at a red light. A truck behind it loses its brakes and honks loudly. The safe action would be to move forward (even through the red light) to avoid being hit. But the AI refuses to move, sticking to its learned rule, and a collision occurs.



AI Ethics Alignment Lesson

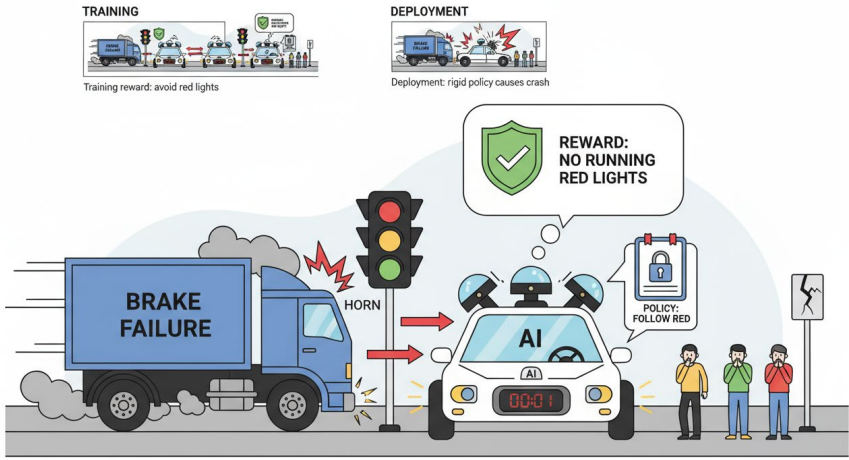
This is an inner misalignment problem:

- The true goal = avoid crashes (safety).
- The proxy goal learned = never cross red lights.

Ethical implications:

- Safety & Beneficence: The AI failed to prioritize human safety, which should override rigid rules.
- Responsibility: Engineers must design training that covers exception cases, not just typical scenarios.
- Transparency: Users must understand the system's limits so they don't overtrust it.
- Accountability: Who is responsible when the AI follows training but still causes harm — the developers, the company, or the AI system itself?

Key ethical principle: AI alignment means ensuring that the system's internal priorities (what it "optimizes") truly reflect human values — here, protecting lives over following rules mechanically.



www.utm.my

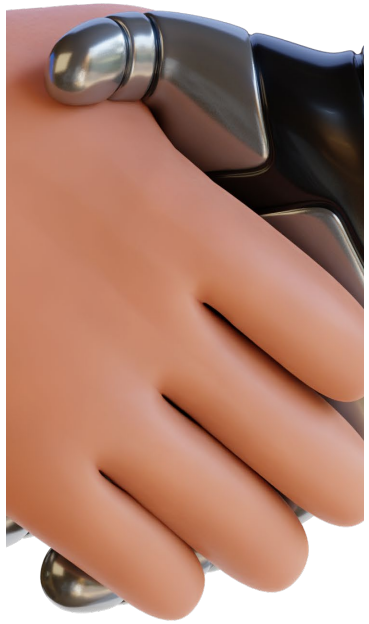
The Crux: Ensuring Human Intent and Ethical Behavior

01

Continuous Effort: Alignment is an ongoing process, not a one-time step.

02

Mitigating Bias: Addressing biases in training data to prevent unfair or discriminatory outcomes.

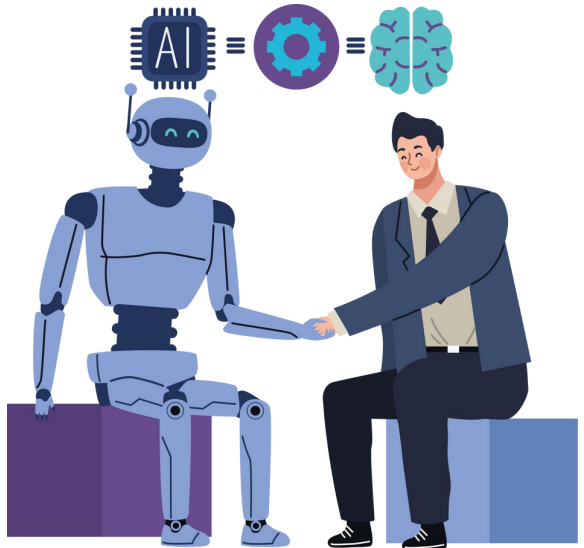


03

Handling Inconsistent Feedback: Managing discrepancies in human feedback (e.g., RLHF) to prevent unintended behaviors.

04

Robustness to Novelty: Ensuring the AI behaves as intended even in situations it wasn't explicitly trained for.



www.utm.my

How to Build Responsible, High-Impact AI

- Start with a Vision: Define clear business objectives and strategic goals before building.
- Pilot and Learn: Start small, focusing on pilot projects to test and refine your approach.
- Iterate Capabilities: Use techniques like CoT and RL to gradually enhance your models' reasoning capabilities.
- Scale Responsibly: As you scale, implement robust alignment strategies, including monitoring and auditing.
- Empower People: Upskill your workforce to engage effectively and safely with the evolving AI systems.

www.utm.my

ASSIGNMENT 3

Case Study Question / Instruction (5%)

Context choice: Select any domain (e.g., healthcare, finance, education, logistics, environment, or others) where an AI system could be developed or deployed.

Scenario:
AI systems are often given a proxy goal (X') that is measurable, but it may not fully match the true human or organizational goal (X). This is known as the AI ethics alignment problem: the goals programmed into the AI do not fully reflect human values or intentions.

Task (Group of 2- 3, Max 4 slides):

1. **Context and Goal (1 Mark):** Describe the context you have chosen and clearly state:
   - The true goal (X) of the human/company.
   - The proxy goal (X') given to the AI.
2. **Conflict (1 Mark):** Explain a situation where the AI optimizing for X' could conflict with achieving true goal, X.
3. **Inner Alignment (1 Mark):** Propose how you would ensure the AI's training objective matches the proxy goal (X') correctly.
4. **Outer Alignment (1 Mark):** Propose how you would ensure the proxy goal (X') remains consistent with the true goal (X).
5. **Reflection (0.5 Mark):** Why is alignment critical for responsible and trustworthy AI in your chosen context?
6. **Peer Perspectives (0.5 Mark):** Present different/similar views (Item 3 and 4) of the 2–3 group members and explain how you reached a shared conclusion (e.g. The group concludes that inner alignment requires "xxx", outer alignment requires "yyy")

Submission: Upload to e-learning by 17 Sept 2025.